The Effect of DRM Paradigm Presentation Modality on False Memory Formation

Psych 121: Laboratory in Cognitive Psychology Final Paper

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Introduction

The topic we are researching is false memory formation and how different presentation modalities influence the likelihood of false memories occurring. False memories are an interesting area of study because they demonstrate that the brain often organizes memories around shared meanings and concepts rather than relying solely on the specific details of an event. Understanding how false memories develop also has important real-world implications. For example, eyewitnesses may unintentionally recall inaccurate details of a crime, or false memories can lead to the formation and spread of misinformation through social media.

False memories have been widely studied using the Deese-Roediger-McDermott (DRM) paradigm, an experimental design in which participants first study a list of semantically related words and then complete a memory test to identify or recall the words from the list (Pardilla-Delgado & Payne, 2017). Findings from this experiment consistently show that participants recognize words that were never presented but are closely related in meaning to the studied words. This occurs because the brain tends to encode the overall meaning of information instead of its exact details.

Existing literature has examined how different presentation modalities influence the formation of false memories. For example, Robin and Mahé (2015) found that participants instructed to generate mental images of words produced fewer false memories than those instructed to generate verbal characteristics of the same words. Their study assessed false memories using both free recall and recognition tests. Wang et al. (2018) further explored this relationship between visual stimuli and false memories by investigating whether participants recognized blurry photographs more quickly when they were associated with false memories. Using a recognition test, they found no significant difference, which suggests that false memories for pictures are driven by conceptual rather than perceptual processing. Smith and Hunt (1998) compared auditory and visual presentation modalities and found that participants who heard word lists produced significantly more false memories than those who read them. These effects were observed on both free recall and recognition memory tests, indicating that visually presented words create more distinctive memory representations than auditory presentations. Similarly, Israel and Schacter (1997) compared picture encoding with word encoding and found that participants shown pictures made significantly fewer false recognitions than those shown written words. Using a recognition test, their findings suggest that pictures produce more distinctive memories and stronger perceptual representations than just words.

Our goal is to directly compare picture, text, and audio stimuli and their effects on false memory rates, a comparison that has not yet been examined in a single study. Our study uses a recognition memory test to measure false memories. This allows for direct comparison with prior research that used recognition tests while extending it to include all three presentation modalities. We hypothesize that the picture + word condition will produce the fewest false memories, followed by the text condition, with the audio condition producing the highest rate of false memories. This prediction aligns with previous research suggesting that pictures create the richest and most distinctive memory representations, while auditory information produces less distinctive memories. If our results differ from this pattern, it would suggest that other factors beyond the richness of perceptual encoding may play a larger role in the formation of false memories.

Methods

Participants

Participants were **30 undergraduate students** enrolled in UCLA Psych 121. All participants were at least 18 years old and provided informed consent before participating in the study. Before beginning the experiment, participants completed a brief demographic questionnaire that collected information about their age, gender, and whether English was their native language. Participants received course credit for their participation.

Design

This study used a **1 × 3 within-subjects (repeated-measures) design**. The independent variable was **presentation modality**, which had three levels: **Written Word, Spoken Word, and Picture + Word**. The dependent variable was **false memory rate**. A false memory was defined as responding **"Yes"** to a critical lure that had not been presented during the study phase. False memory rate was calculated as the proportion of **"Yes"** responses to the critical lure in each presentation modality. All participants experienced all three presentation modalities. To prevent participants from studying the same DRM list more than once, **Latin Square counterbalancing** was used. This ensured that the three DRM word lists (**Sleep, Doctor, and Cold**) were presented equally across the three presentation modalities.

Materials

The experiment used standardized **Deese–Roediger–McDermott (DRM)** word lists, which are commonly used in false memory research. Three DRM lists were used: **Sleep, Doctor, and Cold**. Each list contained 15 semantically related words. Each list also included one critical lure that was strongly associated with the list but was never presented during the study phase. For example, the **Sleep** list included words such as *bed, dream,* and *nap*, but the word *sleep* itself was not presented. The same DRM lists were presented in three different formats: **Written Word**, in which words were presented as text on the screen; **Spoken Word**, in which words were presented as audio recordings; and **Picture + Word**, in which a picture and its corresponding word were presented simultaneously. After each DRM list, participants completed a recognition memory test consisting of **12 words**: **4 previously studied words, 1 critical lure, and 7 new foil words** selected from DRM lists that had not been studied. The experiment was created and administered using **PsychoPy**, which presented the stimuli and automatically recorded participants' responses and reaction times.

Procedure

At the beginning of the experiment, participants read the instructions and provided informed consent. They then completed a demographic questionnaire before beginning the experimental task. PsychoPy assigned each participant to one of three Latin Square counterbalancing orders. Each participant experienced all three presentation modalities, but each DRM list appeared in only one modality. For example, one participant studied the Sleep list in the Written Word condition, the Doctor list in the Spoken Word condition, and the Cold list in the Picture + Word condition, whereas another participant experienced a different combination.

For each condition, participants studied one DRM list containing 15 words presented one at a time. After studying the list, participants completed a 30-second backward counting task, during which they counted backward by threes from a given number. This task was included to reduce the influence of short-term memory before the recognition test. Participants then completed a recognition memory test and responded “Yes” if they remembered studying the word or “No” if they did not.

This study–counting–recognition sequence was repeated three times, once for each presentation modality (Written Word, Spoken Word, and Picture + Word), with a different DRM list used in each condition. Thus, each participant completed all three modalities and studied all three DRM lists, without seeing the same list in more than one modality.

PsychoPy automatically recorded participants’ responses and reaction times. After data collection, false memory rates for each presentation modality were calculated and compared using a one-way repeated-measures ANOVA.

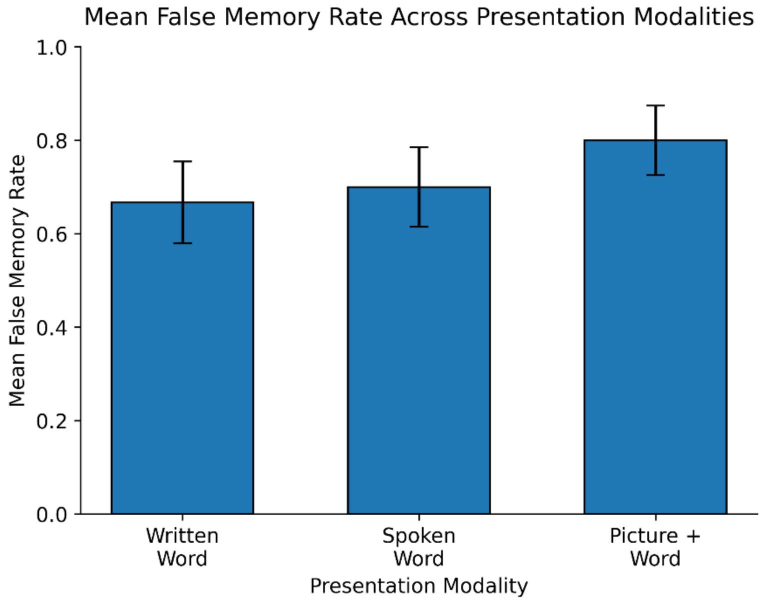
Results

A one-way repeated-measures ANOVA was conducted to examine whether presentation modality affected false memory rates. Mauchly's test indicated that the assumption of sphericity was met ($W = .915$, $p = .288$); therefore, no correction was applied.

The results showed that presentation modality did not significantly affect false memory rates, $F(2, 58) = 1.09$, $p = .344$, $\eta^2 = .016$.

The mean false memory rate was highest in the Picture + Word condition ($M = 0.80$, $SD = 0.41$), followed by the Spoken Word condition ($M = 0.70$, $SD = 0.47$), and the Written Word condition ($M = 0.67$, $SD = 0.48$). However, post hoc comparisons using the Holm correction indicated that none of the pairwise differences were statistically significant. The comparisons between Written Word and Spoken Word (p = .745), Written Word and Picture + Word (p = .310), and Spoken Word and Picture + Word (p = .651) were all non-significant.

Figure 1 shows the mean false memory rate for each presentation modality.



Note. *Error bars represent ±1 standard error of the mean (SEM). No significant differences were observed among the three presentation modalities (p > .05).*

Discussion

The results of this experiment show no significant effect of presentation modality on false memory rates, with the higher false memory rates being slightly higher in the Picture + Word condition, followed by the spoken word condition and finally the written word condition. This differs from our hypothesis of the Picture + Word condition producing the lowest false memory rate, followed by written and then spoken word. The repeated measures ANOVA produced a p-value of .344, suggesting that there is no significant difference between the three conditions.

These findings differ from the previous literature studying false memory rates that found lower false memory rates for visually presented words compared with auditory presentation (Smith & Hunt, 1998) and lower false recognition following picture encoding compared with word encoding (Israel & Schacter, 1997). However, because none of our pairwise comparisons were statistically significant, our results should be viewed cautiously because they may represent random variation in our sample rather than meaningful differences in the presentation modality.

An explanation for these findings could be that false memories are driven by thematic associations rather than perceptual characteristics. Participants may still have encoded the overall meaning of words in the written and auditory presentation even though the perceptual information available to them was stronger in the picture + word condition. This would be consistent with the findings of Wang et al. (2018), which suggested that false memories for pictures are based on conceptual rather than perceptual processing.

There are also limitations to this study. The sample size of 30 people may have been too small to produce statistically significant results. A larger sample size could increase the likelihood of finding effects of presentation modality. We also utilized only three DRM word lists, which may limit the generalizability of the findings because different DRM lists may vary in how strongly they elicit false memories. Each presentation modality also included one critical lure, so false memory rates were based on a single observation per condition for each participant. Additionally, participants completed all three conditions in a single session, which may have led to fatigue or practice effects despite counterbalancing. Future research should include a larger sample size, possibly more word lists, more critical lures, and examine whether other factors like image complexity or emotional context interact with presentation modality to influence false memory formation. Future studies may also compare presentation modality in paired experiments, such as pictures vs words or words vs audio. This may provide a clearer understanding of the specific effects of each presentation modality. Further research could also compare recognition to free recall tasks to see if presentation modality affects participants differently based on the type of memory test they do.

Overall, the study did not find evidence that presentation modality significantly affects participants' false memory rates in the DRM paradigm. These findings suggest that semantic relationships between the studied words may play a greater role in false memory formation than their presentation modality.

References

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